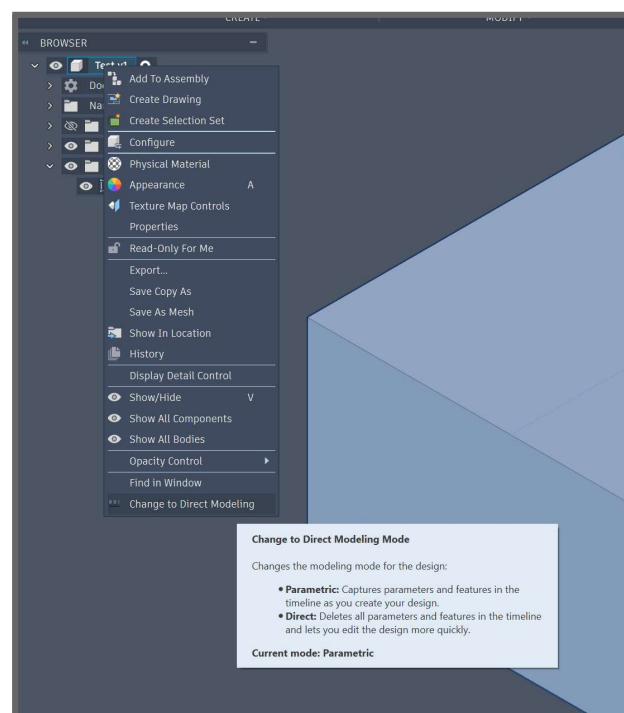


3D Printing and Modeling

Homework #2

1. General rules

1. Homework must be uploaded on Teams.
2. You must upload both the *.stl* export and *.f3d* project file.
3. Turn on *capture history* when 3D modeling in Fusion!
4. When designing, keep in mind that the object needs to be 3D printed.
5. Always try to keep your design as simple as possible.
6. Name *all* of your sketches, bodies, parts, joints accordingly.
7. Constrain your sketches accordingly. Use construction lines when needed.
8. Turn on **parametric modeling** so capture history is enabled!
9. You will be penalised 1 point (out of 10) for each late day.

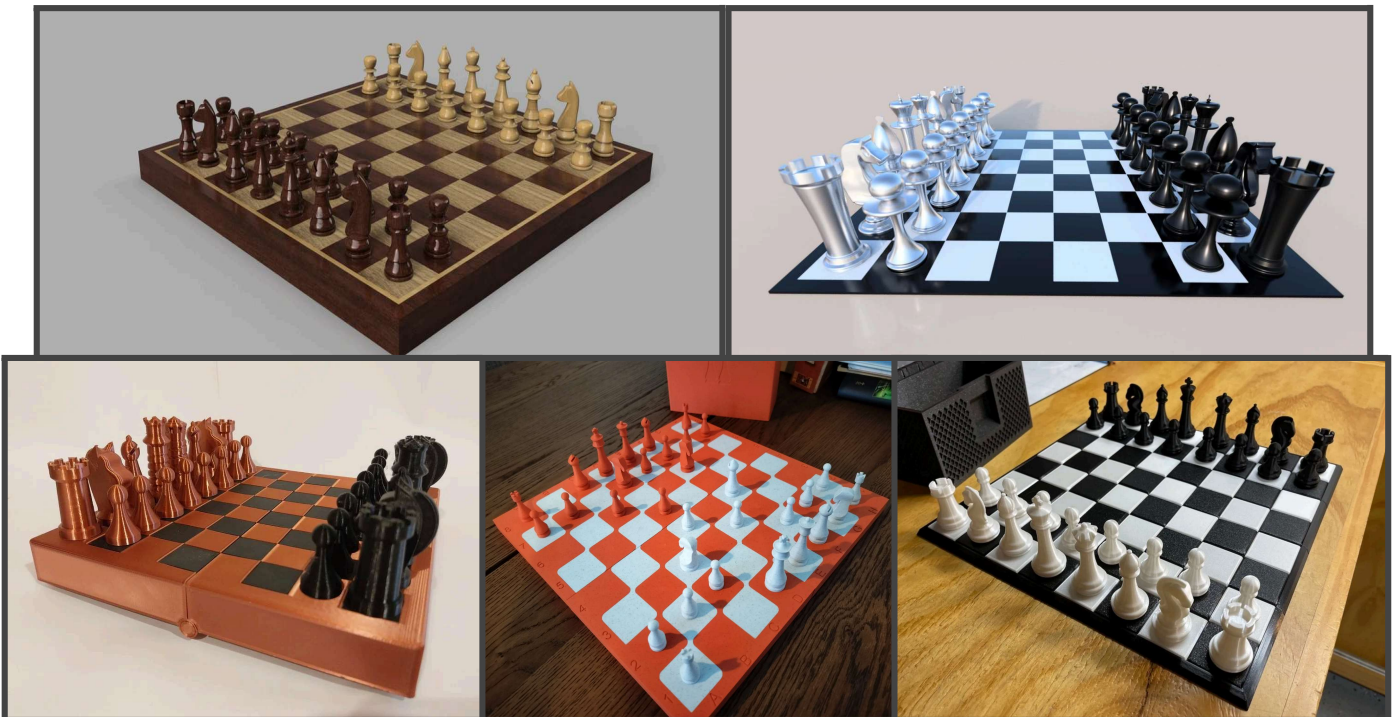


2. Homework assignment

Chess Set

Homework #2

For this homework you must design a fully functional chess board and arrange the pieces correctly. You do not have to print your design, but keep in mind that it still needs to be 3D printable. From now on, you will be penalised if your design is very difficult to print.



Examples

To design this easily and in a time efficient manner you will need to use **Rectangular / Circular Pattern**, **Revolve** and **Sweep** to shape your parts. We expect you to use each one at least once.

Requirements (what you need to upload):

1. **The F3D Fusion 360 project file** containing:
 - a. A chess board, between 40 and 50 cm in each dimension.
 - b. A full chess set (32 pieces, 16 for each color), and a unique design for each of the following:
 - i. A pawn (must be the shortest piece)
 - ii. A rook
 - iii. A knight
 - iv. A bishop
 - v. A king
 - vi. A queen
 - c. Make sure your design is 3D printable.
 - d. All pieces need to fit within one square on the board.
 - e. You will need to duplicate the pieces created and place them on the board. Your design needs to contain at least 33 bodies (32 pieces and one board).
 - f. For a perfect grade you will also need to change the appearance of your designs. Go to **Modify > Appearance** or **Modify > Physical Material**. Make the opposing sides different colors. These were discussed very briefly during the last course. You can search for more information online if you need.
 - g. You can use **Circular Pattern** / **Sweep** to add details to the design of your pieces / board.
2. An **STL export** of your file.

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